

ABDELRHMAN AKRAM IBRAHIM MOHAMED

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PROFESSIONAL SUMMARY

Analytical Computer Science undergraduate specializing in Data Analytics. Skilled in Microsoft Excel, Power BI, SQL, Python, and R to architect end-to-end ETL pipelines, conduct exploratory data analysis, build executive dashboards, and deploy predictive models. Experienced in optimizing reporting workflows, identifying key business metrics, and translating complex data into actionable strategic insights.

EDUCATION

Bachelor of Science in Computer Science (Cybersecurity Major) - Alexandria University

Expected 2028

Relevant Coursework: Data Mining, Machine Learning, Database Management Systems, Data Structures & Algorithms

TECHNICAL SKILLS

Business Intelligence & Reporting: Power BI (DAX, Power Query), Advanced Excel (Pivot Tables, Array Formulas), R Shiny.

Database Management & Querying: SQL (DataCamp Certified), Relational Schema Normalization, Joins, Aggregations, Subqueries.

Programming Languages & Libraries: Python (Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn), R, C++, Java, HTML5, CSS3, JavaScript.

Analytics Methods: Exploratory Data Analysis (EDA), Data Preprocessing, Data Cleaning, Feature Engineering, Clustering, Regression.

PROJECTS

Superstore Interactive Sales Dashboard (Excel)

- Engineered an interactive Excel dashboard analyzing 9,000+ transactional records, tracking 12+ critical KPIs across regional sales and profit margins.
- Leveraged PivotTables, calculated fields, and interactive slicers, reducing ad-hoc reporting time by 40% for business evaluation.

Central Region Superstore BI Analytics (Power BI)

- Architected a comprehensive Power BI dashboard analyzing 3,000+ regional orders, uncovering top product category trends across 13 states.
- Implemented spatial performance analytics and drill-through pages, increasing regional reporting granularity and query accuracy by 25%.

Sales Performance & Customer Behavior Analysis (Power BI)

- Constructed a multi-page Power BI report integrating 4 transactional tables to track revenue growth, product KPIs, and customer acquisition metrics.
- Developed 15+ custom DAX measures and data models to optimize cross-filtering efficiency, accelerating dashboard query execution speed by 30%.

Customer Behavioral Analytics & Executive Dashboard (R, Shiny, Excel)

- Engineered an interactive executive dashboard using R Shiny and Excel, analyzing transactional trends across 5,000+ customer records.
- Executed rigorous ETL preprocessing, including missing value imputation, deduplication, and outlier filtering to achieve 100% data integrity.
- Accelerated executive reporting workflows by 35% by implementing automated visualization parameters and interactive filter controls.
- Applied K-Means clustering across 4 customer segments and leveraged Apriori association algorithms to identify high-converting cross-selling opportunities.

Student Performance Prediction & Clustering System (Python, SQL)

- Queried 1,000+ student academic records using SQL and Python (Pandas) to extract 10+ performance drivers and benchmark student trajectories.
- Developed K-Medoids clustering algorithms integrated with Fuzzy Logic rules, segmenting performance tiers with 88% classification alignment.
- Implemented Genetic Algorithms for automated feature selection, enhancing model prediction efficiency and interpretability by 20%.

Automotive Market Pricing & Predictive Analytics (Python, Scikit-Learn)

- Executed exploratory data analysis (EDA) on 10,000+ vehicle market entries in Python to quantify key price valuation drivers across 8 attributes.
- Transformed raw attributes using categorical encoding, robust feature scaling, and outlier handling to maximize predictive model readiness.
- Optimized ensemble classification and regression algorithms via 5-fold cross-validation, improving target valuation accuracy by 18%.

FIFA Player Market Value Regression Modeling (Python, Ridge/Lasso)

- Processed high-dimensional athletic datasets of 18,000+ players using Python, executing one-hot encoding across 15+ athletic performance metrics.
- Evaluated Linear, Lasso, and Ridge regression models, establishing Ridge Regression as optimal with a 12% improvement in R² score.

CERTIFICATIONS

- Data Fundamentals** – IBM SkillsBuild (2026)
- Intermediate SQL Certification** – DataCamp (2026)
- Introduction to SQL Certification** – DataCamp (2026)
- Introduction to Python Certification** – DataCamp (2026)
- Python Programming Basics** – ITI Mahara Tech (2025)

LANGUAGES

- Arabic:** Native | **English:** B2 Upper-Intermediate (Fluent Spoken & Technical)